

Problem 1:

```
Ch7_PrExercise5.java
1- import java.util.Scanner;
2
3- public class Ch7_PrExercise5 {
4
5     // one Scanner that the whole class can share
6     private static final Scanner console = new Scanner(System.in);
7
8     public static void main(String[] args) {
9
10         int choice;           // menu selection
11         double length, width;  // for rectangle
12         double radius, height; // for circle / cylinder
13
14         // ----- main loop -----
15         do {
16             // 1. show menu
17             System.out.println();
18             System.out.println("This program can calculate ...");
19             System.out.println(" 1 : Area of a rectangle");
20             System.out.println(" 2 : Area of a circle");
21             System.out.println(" 3 : Volume of a cylinder");
22             System.out.println("-1 : Exit");
23             System.out.print("Enter your choice: ");
24             choice = console.nextInt();
25             System.out.println();
26
27             // 2. act on choice
28             switch (choice) {
29
30                 case 1: // rectangle
31                     System.out.print("Enter the length of the rectangle: ");
32                     length = console.nextDouble();
33                     System.out.print("Enter the width of the rectangle: ");
34                     width = console.nextDouble();
35                     System.out.printf("Area = %.2f\n\n",
36                                     rectangleArea(length, width));
37                     break;
38
39                 case 2: // circle
40                     System.out.print("Enter the radius of the circle: ");
41                     radius = console.nextDouble();
42                     System.out.printf("Area = %.2f\n\n",
43                                     circleArea(radius));
44                     break;
45
46             }
```

```

39         case 2: // circle
40             System.out.print("Enter the radius of the circle: ");
41             radius = console.nextDouble();
42             System.out.printf("Area = %.2f\n\n",
43                 circleArea(radius));
44             break;
45
46         case 3: // cylinder
47             System.out.print("Enter the radius of the cylinder base: ");
48             radius = console.nextDouble();
49             System.out.print("Enter the height of the cylinder: ");
50             height = console.nextDouble();
51             System.out.printf("Volume = %.2f\n\n",
52                 cylinderVolume(radius, height));
53             break;
54
55         case -1: // quit
56             System.out.println("Program terminated – goodbye!");
57             break;
58
59         default: // invalid entry
60             System.out.println("Invalid choice – please try again.\n");
61     }
62
63     } while (choice != -1);
64 }
65
66 /* ----- helper methods ----- */
67
68 /** area of a rectangle */
69 public static double rectangleArea(double l, double w) {
70     return l * w;
71 }
72
73 /** area of a circle */
74 public static double circleArea(double r) {
75     return Math.PI * r * r;
76 }
77
78 /** volume of a right circular cylinder */
79 public static double cylinderVolume(double r, double h) {
80     return Math.PI * r * r * h;
81 }
82 }
83

```

Output

```
This program can calculate ?
  1 : Area of a rectangle
  2 : Area of a circle
  3 : Volume of a cylinder
-1 : Exit
Enter your choice: 1

Enter the length of the rectangle: 13
Enter the width of the rectangle: 12
Area = 156.00

This program can calculate ?
  1 : Area of a rectangle
  2 : Area of a circle
  3 : Volume of a cylinder
-1 : Exit
Enter your choice: 2

Enter the radius of the circle: 14
Area = 615.75

|
This program can calculate ?
  1 : Area of a rectangle
  2 : Area of a circle
  3 : Volume of a cylinder
-1 : Exit
Enter your choice: 3

Enter the radius of the cylinder base: 10
Enter the height of the cylinder: 15
Volume = 4712.39

This program can calculate ?
  1 : Area of a rectangle
  2 : Area of a circle
  3 : Volume of a cylinder
-1 : Exit
Enter your choice: -1

Program terminated ? goodbye!
```

Problem 2:

Ch7_PrExercise5_Problem2.java

```
1 import java.util.Scanner;
2
3 public class Ch7_PrExercise5_Problem2 {
4
5     /*----- constants & tools -----*/
6     private static final double PI = 3.1416; // use the given value
7     private static final Scanner CONSOLE = new Scanner(System.in);
8
9     /*----- main driver -----*/
10    public static void main(String[] args) {
11
12        // 1. read two points: the centre (x1,y1) and a point on the circle (x2,y2)
13        System.out.print("Enter the centre of the circle (x1 y1): ");
14        double x1 = CONSOLE.nextDouble();
15        double y1 = CONSOLE.nextDouble();
16
17        System.out.print("Enter a point on the circle (x2 y2): ");
18        double x2 = CONSOLE.nextDouble();
19        double y2 = CONSOLE.nextDouble();
20        System.out.println();
21
22        // 2. compute everything
23        double r = radius(x1, y1, x2, y2);
24        double diameter = 2 * r;
25        double circ = circumference(r);
26        double circleArea = area(r);
27
28        // 3. display results
29        System.out.printf("Radius      : %.4f\n", r);
30        System.out.printf("Diameter   : %.4f\n", diameter);
31        System.out.printf("Circumference : %.4f\n", circ);
32        System.out.printf("Area       : %.4f\n", circleArea);
33    }
34
35    /*----- required methods -----*/
36
37    /** distance between two points (x1, y1) and (x2, y2) */
38    public static double distance(double x1, double y1, double x2, double y2) {
39        return Math.sqrt(Math.pow(x2 - x1, 2) + Math.pow(y2 - y1, 2));
40    }
41
42    /** radius of the circle given its centre and a boundary point */
43    public static double radius(double x1, double y1, double x2, double y2) {
44        return distance(x1, y1, x2, y2); // simply the distance
45    }
46
47    /** circumference of a circle with radius r */
48    public static double circumference(double r) {
49        return 2 * PI * r;
50    }
51
52    /** area of a circle with radius r */
53    public static double area(double r) {
54        return PI * r * r;
55    }
56 }
```

Output

```
Enter the centre of the circle (x1 y1): 5
10
Enter a point on the circle (x2 y2): 10
5
```

```
Radius      : 7.0711
Diameter    : 14.1421
Circumference : 44.4289
Area        : 157.0800
```

```
=== Code Execution Successful ===
```

Problem 3:

```
PalindromCheck.java    Share  Run

1 import java.util.Scanner;
2
3 public class PalindromCheck {
4
5     public static void main(String[] args) {
6
7         Scanner in = new Scanner(System.in);
8         int n;
9
10        System.out.print("Enter a non-negative integer (-1 to quit): ");
11        while ((n = in.nextInt()) != -1) {
12            System.out.printf("%d is %sa palindrome.%n",
13                               n,
14                               isPalindrome(n) ? "" : "not ");
15            System.out.print("\nEnter another number (-1 to quit): ");
16        }
17        System.out.println("Done!");
18    }
19
20    /**
21     * Returns true if the given non-negative integer is a palindrome.
22     * Works purely with arithmetic - no String conversion.
23     */
24    public static boolean isPalindrome(int n) {
25        if (n < 10) return true; // any single digit qualifies
26
27        int original = n;
28        int reversed = 0;
29
30        // build the reverse of n, digit by digit
31        while (n > 0) {
32            reversed = reversed * 10 + n % 10;
33            n /= 10;
34        }
35        return original == reversed;
36    }
37 }
38
```

Output

Enter a non-negative integer (-1 to quit): 10
10 is not a palindrome.

Enter another number (-1 to quit): 2
2 is a palindrome.

Enter another number (-1 to quit): 3
3 is a palindrome.

Enter another number (-1 to quit): 4
4 is a palindrome.

Enter another number (-1 to quit): 5
5 is a palindrome.

Enter another number (-1 to quit): 6
6 is a palindrome.

Enter another number (-1 to quit): 8
8 is a palindrome.

Enter another number (-1 to quit): 9
9 is a palindrome.

Enter another number (-1 to quit): 12
12 is not a palindrome.

Enter another number (-1 to quit): 11
11 is a palindrome.

Enter another number (-1 to quit): |

Problem 4:

```
MeanStdDev.java
1- import java.util.Scanner;
2
3- public class MeanStdDev {
4
5-     // ----- helper methods -----
6
7-     /** returns the arithmetic mean of the five numbers */
8-     public static double mean(double[] a) {
9-         double sum = 0;
10-         for (double v : a) sum += v;
11-         return sum / a.length;           // here a.length == 5
12-     }
13
14-     /** returns the population SD given the data and its mean */
15-     public static double stdDev(double[] a, double x̄) {
16-         double sumSq = 0;
17-         for (double v : a) sumSq += Math.pow(v - x̄, 2);
18-         return Math.sqrt(sumSq / a.length); // divide by 5 as specified
19-     }
20
21-     // ----- main driver -----
22
23-     public static void main(String[] args) {
24
25-         Scanner in = new Scanner(System.in);
26-         double[] num = new double[5];
27
28-         System.out.println("Enter five numbers:");
29-         for (int i = 0; i < 5; i++) {
30-             System.out.printf(" #d: ", i + 1);
31-             num[i] = in.nextDouble();
32-         }
33-         System.out.println();
34
35-         double avg = mean(num);
36-         double sd = stdDev(num, avg);
37
38-         System.out.printf("Mean           = %.4f\n", avg);
39-         System.out.printf("Standard Deviation = %.4f\n", sd);
40-     }
41 }
```

Output

Enter five numbers:

#1: 10

#2: 50

#3: 30

#4: 40

#5: 25

Mean = 31.0000

Standard Deviation = 13.5647

=== Code Execution Successful ===

Problem 5:

FitnessCenterQuote.java

[illegible]


```

46     }
47     } while (choice != -1);
48 }
49
50 /* =====
51 | | | | | method 1: display general info
52 | | | | | =====*/
53 private static void showInfo() {
54     System.out.println("=== Fitness-Center Pricing & Discounts ===");
55     System.out.printf("Monthly membership fee      : $%.2f%n", MONTHLY_RATE);
56     System.out.printf("Personal-training session fee : $%.2f%n", SESSION_RATE);
57     System.out.println("\nDiscounts that may apply:");
58     System.out.println("    * Senior citizen          30 % off membership portion");
59     System.out.println("    * Pay ≥ 12 months up-front  15 % off membership portion");
60     System.out.println("    * Buy > 5 training sessions  20 % off each session");
61     System.out.println("Only one membership discount (senior OR long-term) can apply.");
62     System.out.println("Training-session discount is independent.");
63     System.out.println("=====\\n");
64 }
65
66 /* =====
67 | | | | | method 2: collect all data from the customer
68 | | | | | =====*/
69 private static CustomerInfo collectInput() {
70
71     CustomerInfo c = new CustomerInfo();
72
73     System.out.print("How many months of membership? : ");
74     c.months = CONSOLE.nextInt();
75
76     System.out.print("How many personal-training sessions? : ");
77     c.sessions = CONSOLE.nextInt();
78
79     System.out.print("Is the customer a senior citizen (Y/N)? : ");
80     char ans = CONSOLE.next().trim().toUpperCase().charAt(0);
81     c.senior = (ans == 'Y');
82
83     System.out.println();
84     return c;
85 }
86
87 /* =====
88 | | | | | method 3: compute the membership cost
89 | | | | | =====*/
90 private static double calcCost(CustomerInfo c) {
91
92     /* -- membership cost ----- */
93     double membershipSubtotal = c.months * MONTHLY_RATE;
94
95     // apply *one* of the membership discounts if applicable
96     if (c.senior) {
97         membershipSubtotal *= (1 - DISC_SENIOR); // senior beats long-term
98     } else if (c.months >= 12) {
99         membershipSubtotal *= (1 - DISC_LONG_TERM);
100     }
101
102     /* -- personal-training cost ----- */
103     double sessionSubtotal = c.sessions * SESSION_RATE;
104
105     if (c.sessions > 5) {
106         sessionSubtotal *= (1 - DISC_SESSIONS);
107     }
108
109     /* -- grand total ----- */
110     return membershipSubtotal + sessionSubtotal;
111 }
112 }
113

```

Output

Clear

```
*** Main Menu ***
1 ? Display price / discount information
2 ? Quote a new membership
-1 ? Exit
Enter choice: 1

=== Fitness-Center Pricing & Discounts ===
Monthly membership fee      : $50.00
Personal-training session fee : $30.00

Discounts that may apply:
? Senior citizen            30 % off membership portion
? Pay ? 12 months up-front  15 % off membership portion
? Buy > 5 training sessions  20 % off each session
Only one membership discount (senior OR long-term) can apply.
Training-session discount is independent.
=====

*** Main Menu ***
1 ? Display price / discount information
2 ? Quote a new membership
-1 ? Exit
Enter choice: 2

How many months of membership? : 3
How many personal-training sessions? : 1
Is the customer a senior citizen (Y/N)? : N

Total cost for this membership: $180.00
```

```
*** Main Menu ***
1 ? Display price / discount information
2 ? Quote a new membership
-1 ? Exit
Enter choice: 2

How many months of membership? : 4
How many personal-training sessions? : 5
Is the customer a senior citizen (Y/N)? : Y

Total cost for this membership: $290.00

*** Main Menu ***
1 ? Display price / discount information
2 ? Quote a new membership
-1 ? Exit
Enter choice: -1

Good-bye!

=== Code Execution Successful ===
```

Problem 6:

```
BoxOptimizer.java
1 import java.util.Scanner;
2
3 public class BoxOptimizer {
4
5     /* ----- helper class to bundle the results ----- */
6     private static class Optimum {
7         double x; // best cut size
8         double volume; // maximum volume
9     }
10
11     /* =====
12     | Method that SATISFIES the specification:
13     | → takes length & width, returns x-opt and V-max
14     | =====*/
15     public static Optimum bestCut(double L, double W) {
16
17         // Solve  $12x^2 - 4(L+W)x + LW = 0$  → see derivation above
18         double a = 12.0, b = -4.0 * (L + W), c = L * W;
19         double disc = b * b - 4 * a * c; // guaranteed ≥ 0
20         double sqrtDisc = Math.sqrt(disc);
21
22         double x1 = (-b - sqrtDisc) / (2 * a); // smaller root
23         // (The larger root is always ≥ min(L,W)/2 and is useless.)
24
25         Optimum out = new Optimum();
26         out.x = x1;
27         out.volume = x1 * (L - 2 * x1) * (W - 2 * x1);
28         return out;
29     }
30
31     /* =====
32     | | | | | main program
33     | =====*/
34     public static void main(String[] args) {
35
36         Scanner in = new Scanner(System.in);
37         System.out.print("Enter the area of the cardboard (square inches): ");
38         double area = in.nextDouble();
39
40         /* Best rectangle for fixed area is a square */
41         double side = Math.sqrt(area); // L = W = √A
42
43         Optimum opt = bestCut(side, side);
44
45         /* ----- display ----- */
46         System.out.printf("%nOptimal cardboard dimensions:%n");
47         System.out.printf(" Length (L) : %.3f in%n", side);
48         System.out.printf(" Width (W) : %.3f in%n", side);
49         System.out.printf("%nCut a square of side x = %.3f in from each corner.%n",
50             opt.x);
51         System.out.printf("Maximum box volume = %.3f cubic inches%n",
52             opt.volume);
53     }
54 }
55
```

Output

Enter the area of the cardboard (square inches): 70

Optimal cardboard dimensions:

Length (L) : 8.367 in

Width (W) : 8.367 in

Cut a square of side x = 1.394 in from each corner.

Maximum box volume = 43.382 cubic inches

=== Code Execution Successful ===

Problem 7:

```
PowerLineOptimizer.java
1 import java.util.Scanner;
2
3 public class PowerLineOptimizer {
4
5     private static final Scanner in = new Scanner(System.in);
6
7     /* -----
8      | Method that returns: {xOpt, underwaterLen, landLen, cost}
9      | -----*/
10    private static double[] solve(double W, double D, double cU, double cL) {
11
12        double xOpt;
13
14        if (cU <= cL) { // underwater not dearer => all UW
15            xOpt = D;
16        } else {
17            xOpt = (cL / Math.sqrt(cU*cU - cL*cL)) * W; // calculus result
18            if (xOpt > D) xOpt = D; // cannot land beyond the factory
19            if (xOpt < 0) xOpt = 0; // numeric safety
20        }
21
22        double lenUW = Math.hypot(W, xOpt); // sqrt(W^2 + x^2)
23        double lenLD = D - xOpt; // remaining overland
24        double cost = cU * lenUW + cL * lenLD;
25
26        return new double[] { xOpt, lenUW, lenLD, cost };
27    }
28
29    /* -----*/
30    public static void main(String[] args) {
31
32        System.out.print("Width of the river (ft): ");
33        double W = in.nextDouble();
34
35        System.out.print("Down-river distance to the factory (ft): ");
36        double D = in.nextDouble();
37
38        System.out.print("Cost per foot UNDERWATER ($): ");
39        double cU = in.nextDouble();
40
41        System.out.print("Cost per foot OVER LAND ($): ");
42        double cL = in.nextDouble();
43
44        double[] ans = solve(W, D, cU, cL);
45
46        System.out.printf("\nOptimal landing point is x = %.2f ft downstream.\n",
47            ans[0]);
48        System.out.printf(" Underwater length : %.2f ft\n", ans[1]);
49        System.out.printf(" Over-land length : %.2f ft\n", ans[2]);
50        System.out.printf("Minimum total cost : $%.2f\n", ans[3]);
51    }
52 }
53
```

Output

```
Width of the river (ft): 2640
Down-river distance to the factory (ft): 42240
Cost per foot UNDERWATER ($): 9
Cost per foot OVER LAND ($): 7

Optimal landing point is x = 3266.83 ft downstream.
 Underwater length : 4200.21 ft
 Over-land length : 38973.17 ft
Minimum total cost : $310614.10
```

```
=== Code Execution Successful ===
```

Problem 8:

```
PipeCorner.java
1 import java.util.Scanner;
2
3 public class PipeCorner {
4
5     /** returns  $(a^{2/3} + b^{2/3})^{3/2}$  */
6     public static double longestPipe(double a, double b) {
7         double a23 = Math.pow(a, 2.0 / 3.0);
8         double b23 = Math.pow(b, 2.0 / 3.0);
9         return Math.pow(a23 + b23, 1.5);    // 3/2 power
10    }
11
12    public static void main(String[] args) {
13
14        Scanner in = new Scanner(System.in);
15
16        System.out.print("Width of the first corridor (ft): ");
17        double a = in.nextDouble();
18
19        System.out.print("Width of the second corridor (ft): ");
20        double b = in.nextDouble();
21
22        double L = longestPipe(a, b);
23        long Lround = Math.round(L);    // nearest foot
24
25        System.out.printf("Longest pipe that fits = %.2f ft\n", L);
26        System.out.printf("Rounded to nearest foot = %d ft\n", Lround);
27    }
28 }
29
```

Output

```
Width of the first corridor (ft): 8
Width of the second corridor (ft): 5

Longest pipe that fits = 18.22 ft
Rounded to nearest foot = 18 ft

=== Code Execution Successful ===
```